

# From Concept to Code: A Master Directive for Building Project Oracle, an Open-Source Institutional Options Analytics Platform

## Architectural Blueprint for a Low-Latency Microstructure Engine

The construction of "Project Oracle," a self-hosted options analytics terminal designed to rival proprietary systems like Skylit, necessitates an architectural foundation built upon three non-negotiable pillars: real-time data ingestion, high-performance analytical processing, and a flexible, columnar OLAP storage system. This blueprint rejects conventional approaches in favor of a specialized, low-latency stack optimized for the computational demands of market microstructure analysis. The entire system is predicated on the principle of minimizing latency at every stage, from the arrival of a new market tick to its transformation into a quantifiable signal. The core architecture is defined by a strict set of constraints and invariant rules that govern the interaction between its constituent parts. First, the system must operate on a "no polling" policy; all real-time state changes must be driven exclusively by asynchronous WebSocket streams from broker APIs, such as Schwab's Streamer service [11](#). This ensures that the system's internal state remains synchronized with the market with minimal delay, avoiding the inefficiencies and inherent lag of periodic REST API calls [1](#). Second, the architecture is "OLAP-first," mandating that all raw tick data and options chain information be written to an in-memory DuckDB instance. This choice eliminates network overhead associated with client-server database architectures and enables sub-millisecond analytical query performance directly against the ingested data [3](#) [122](#). Third, any computation-intensive logic, particularly the calculation of higher-order Greeks and VPIN, must be compiled using Just-In-Time (JIT) compilation via Numba. This transforms Python functions into highly optimized machine code, achieving C-level execution speeds necessary for real-time processing [8](#) [10](#). Finally, all I/O operations, including network communication with the WebSocket server and database writes, must be managed through an `asyncio`-based event loop to maximize concurrency and throughput without blocking the main execution thread [74](#).

The physical implementation of this architecture begins with the creation of a monorepo structure, which logically organizes the distinct components of the application. The proposed directory layout consists of `/ingestion`, `/analytics`, `/api`, and `/ui` modules [74](#). The `/ingestion` module will house the WebSocket client responsible for connecting to the broker, subscribing to Level 2 options and underlying asset ticks, and parsing the incoming JSON messages [11](#). The `/analytics` module will contain the core mathematical kernels, where Numba-compiled functions will perform calculations for GEX, VEX, Vanna, Charm, and VPIN. The `/api` module, built with FastAPI, will expose endpoints for the UI to request processed data or trigger specific analyses. Lastly, the `/ui` module will implement the Dash/Plotly frontend, responsible for rendering the various charts and tables that visualize the analytical output [3](#). For long-term maintenance and future expansion, an additional `/research` directory is recommended to house newly discovered open-source libraries and academic papers that are under consideration for integration [23](#). This modular structure promotes separation of concerns, simplifies testing, and facilitates team collaboration if the project expands beyond a solo developer.

The cornerstone of the data pipeline is the real-time ingestion layer. Instead of relying on batch-oriented REST APIs, the system must establish a persistent WebSocket connection to a broker's streaming API, such as Schwab's [11](#). The reference repository `tylerebowers/Schwabdev` provides a concrete starting point for implementing this component, offering proven logic for managing the WebSocket connection, parsing the complex Level 2 options stream, and handling the automatic refresh of authentication tokens to maintain connectivity [11](#). All incoming data—ticks for the underlying asset (SPY, SPX, QQQ) and full options chains for various strikes and expirations—must be handled asynchronously. To manage the high throughput of this data, a bounded asynchronous queue is employed. Incoming WebSocket messages are placed into this queue, and a dedicated writer task processes the queue by batching messages and executing bulk `INSERT` statements into the DuckDB database. This batching strategy, with a target interval of 50-100 milliseconds, is a critical optimization. Writing each individual tick to the database would introduce prohibitive I/O overhead; instead, bulk inserts are far more efficient and align with DuckDB's strengths in vectorized, columnar data loading [122](#).

Once ingested, the data must be stored in a format conducive to fast, ad-hoc analytical queries. While Pandas DataFrames are a common tool in quantitative analysis, their performance degrades significantly when data size exceeds single-core memory capacity, leading to linear slowdowns [2](#). DuckDB offers a superior alternative for this use case. It is an in-process analytical database that requires no separate server installation, allowing it to run within the same process as the Python application [3](#). Its SQL-native interface

allows for direct querying of various data sources, including Pandas DataFrames, CSV files, and Parquet files, making it incredibly versatile [3](#) . The proposed schema for the in-memory DuckDB instance is designed to capture the essential elements of market microstructure. It includes a `ticks` table to store per-tick data for the underlying asset, with columns for `timestamp`, `symbol`, `bid`, `ask`, `last`, `volume`, and `oi`. A second table, `chains`, is designed to hold the complete options chain snapshot for a given moment, with columns for `timestamp`, `symbol`, `expiry`, `strike`, `type` (Call/Put), `oi`, and the first-order Greeks (`delta`, `gamma`, `theta`, `vega`) [75](#) . A third table, `lob_snapshots`, is also included to capture Limit Order Book (LOB) data, storing top-of-book bid/ask sizes and prices, which is crucial for calculating metrics like Quote Imbalance [126](#). By writing all data to this DuckDB instance, the system creates a unified, time-indexed repository of market state that can be queried with SQL for any analytical purpose, from simple lookups to complex aggregations required for VPIN calculations [3](#) .

The final piece of the architectural puzzle is performance optimization through JIT compilation. As detailed in the subsequent sections, the calculation of dealer gamma exposure and volume-synchronized probability of informed trading (VPIN) involves intensive loops over large datasets. Standard Python implementations of these algorithms would be orders of magnitude too slow for real-time processing. Numba addresses this by compiling Python functions to native machine code at runtime [10](#) . By decorating the core mathematical functions with the `@numba.njit` decorator, the system can achieve near-C speeds, transforming computationally expensive tasks into manageable operations [8](#) . For example, a function that calculates the GEX for an entire options chain of thousands of strikes would iterate over arrays of open interest and gammas. Numba can vectorize this operation, applying the calculation across the entire array in a single, highly optimized CPU instruction set. Furthermore, for functions that are amenable to parallel execution, Numba's `parallel=True` option can automatically distribute the workload across multiple CPU cores, providing an additional performance boost [44](#) [46](#) . The strict adherence to using Numba for all quantitative logic is not merely an optimization but a fundamental requirement for meeting the project's real-time objectives. This combination of asynchronous ingestion, in-process OLAP storage, and JIT-compiled analytics forms a robust, scalable, and high-performance architectural blueprint for Project Oracle.

| Component            | Technology                                | Rationale                                                                                                                                                                      |
|----------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Real-Time Ingestion  | Python, WebSockets, schwab-trader library | Provides a persistent, low-latency data feed from the broker, avoiding polling delays. The library offers proven connection management and parsing logic <a href="#">11</a> .  |
| Database Backend     | DuckDB (in-process, OLAP)                 | Eliminates network latency, supports sub-millisecond analytical queries via SQL, and runs entirely within the application process <a href="#">3</a> <a href="#">122</a> .      |
| Mathematical Kernels | Python, Numba JIT                         | Compiles computationally intensive functions (GEX, VPIN) to near-native speed, enabling real-time calculation on large datasets <a href="#">8</a> <a href="#">10</a> .         |
| Asynchronous I/O     | Python asyncio                            | Manages concurrent network and database operations efficiently, preventing blocking and maximizing resource utilization <a href="#">74</a> .                                   |
| Frontend Framework   | Dash, Plotly                              | Enables rapid development of interactive web-based dashboards and financial charting capabilities <a href="#">3</a> .                                                          |
| CI/CD & Deployment   | GitHub Actions, Docker, GHCR              | Leverages GitHub Student Pack for automated testing, linting, and container registry services to streamline development and deployment <a href="#">70</a> <a href="#">71</a> . |

This architectural blueprint, while demanding in its initial setup, provides the necessary performance ceiling and flexibility to execute the ambitious goals of Project Oracle. It moves beyond retail-grade tools by building a system from the ground up with institutional requirements for speed, accuracy, and analytical depth.

## The Quantitative Core: Implementing Dealer Gamma Exposure and Flow Toxicity

The heart of Project Oracle lies in its ability to calculate and interpret two of the most critical concepts in modern options market microstructure: dealer gamma exposure (GEX) and order flow toxicity, measured by Volume-Synchronized Probability of Informed Trading (VPIN). These metrics provide a window into the structural pressures and hidden risks embedded within the market, moving beyond simple price action to reveal the hedging behavior of market makers and the informational content of trades. The accurate and efficient implementation of these mathematical models is paramount, and they must be realized through the high-performance computing paradigm established by the Numba JIT compiler. The theoretical foundations for these metrics are well-established in academic literature and are actively used by professional traders to gauge market regimes and anticipate volatility shifts [21](#) [22](#) [42](#).

Dealer Gamma Exposure (GEX) is a measure of the total hedging pressure exerted by dealers (market makers) in response to movements in the underlying asset's price. It is derived from the sum of all open option contracts multiplied by their respective gammas

75 . The standard formula for GEX is expressed as  $GEX = \sum (OI_i \times \Gamma_i \times \Delta_i \times 100 \times Spot)$ , where  $OI_i$  is the open interest at strike  $i$ ,  $\Gamma_i$  is the gamma,  $\Delta_i$  is the delta, and  $Spot$  is the current price of the underlying asset 75 . The factor of 100 represents the typical contract multiplier for stock options. A crucial aspect of this calculation is the sign convention: for dealer positioning, call options are typically assigned a positive value and put options a negative value, reflecting the assumption that dealers are short customer positions 41 75 . The resulting GEX figure has profound implications for market dynamics. A positive GEX indicates that dealers are net long gamma, meaning they hedge *against* price moves by selling into rallies and buying into dips, which tends to stabilize the market and compress volatility 75 97 . Conversely, a negative GEX signifies that dealers are net short gamma, causing them to chase price moves by selling into dips and buying into rallies, which amplifies volatility and encourages trending behavior 75 97 . The "Zero Gamma" (ZG) level, the strike price where the cumulative net GEX flips from positive to negative, acts as a powerful structural support or resistance level. Zones of extremely high positive or negative GEX, known as "Gamma Walls," represent areas where dealer hedging activity is so intense that it can halt or reverse price movement 75 . The implementation of GEX calculation must pull the entire options chain (strikes, OI, deltas, gammas) from the DuckDB `chains` table, pass this data as NumPy arrays to a Numba-compiled function, and return a 1D array of GEX values corresponding to each strike price. This vectorized approach is essential for performance 9 .

Flow toxicity, or the presence of informed trading, is quantified by the VPIN metric. Developed by Easley, Lopez de Prado, and O'Hara, VPIN measures the imbalance between aggressive buying and selling volume, providing a real-time indicator of liquidity risk and potential price disruption 22 69 . Unlike traditional volume imbalance which uses fixed time windows, VPIN employs a "volume clock" that divides the total trading volume into equal-sized buckets 68 . This volume-synchronization makes the metric more robust to periods of high or low trading activity. The VPIN algorithm proceeds in several steps. First, the total volume for a period is partitioned into  $N$  buckets, each containing a volume  $V$  43 . Second, the volume within each bucket is classified as either buy or sell aggressiveness. This is achieved using the "Bulk Volume Classification" method, which classifies volume based on the path of the price within the bucket 126. The Buy Volume ( $V_\tau^B$ ) and Sell Volume ( $V_\tau^S$ ) for bucket  $\tau$  are estimated using the properties of a random walk, where the classification depends on the observed price change and an estimate of volatility 43 . Third, the absolute difference between buy and sell volume is summed across all buckets, and this sum is normalized by the total volume to produce the VPIN value:  $VPIN = \frac{\sum_{\tau=1}^n |V_\tau^B - V_\tau^S|}{\sum_{\tau=1}^n V_\tau}$  over a rolling window of  $n$  buckets 68 .

Simply observing the VPIN value is insufficient for generating reliable signals; its true power comes from analyzing its statistical distribution. By calculating the Cumulative Distribution Function (CDF) of the rolling VPIN time series, one can determine the percentile rank of the current VPIN reading. This allows for the generation of z-scores, creating a standardized signal that flags statistically significant deviations from normal flow conditions [29](#) [68](#). The implementation of the VPIN engine requires maintaining a rolling state of volume buckets in memory. New trade data from the WebSocket stream is continuously fed into this state, buckets are filled and processed, and the VPIN value and its CDF/z-score are updated in real-time. Reference implementations found in repositories like `yt-feng/VPIN` and `theopenstreet/VPIN_HFT` provide the necessary algorithmic details for this complex calculation [68](#).

To further enrich the analysis of dealer behavior, the project explicitly requests the implementation of higher-order Greeks, specifically Vanna and Charm. These metrics offer deeper insights into portfolio sensitivity beyond the first-order effects captured by Delta and Gamma.

- **Vanna**, also known as  $D_{vegaDspot}$  or  $D_{deltaDvol}$ , is the second derivative of an option's price with respect to both the underlying spot price ( $S$ ) and implied volatility ( $\sigma$ ):  $Vanna = \frac{d^2V}{dSd\sigma}$  [9](#). It measures how an option's delta changes as the underlying asset's price moves, or equivalently, how its vega changes as the price changes. Vanna is a critical risk metric for portfolios with exposure to both directional moves and volatility shifts.
- **Charm**, also known as  $D_{eltasDtime}$ , is the second derivative of an option's price with respect to the underlying spot price ( $S$ ) and time ( $t$ ):  $Charm = \frac{d^2V}{dSdt}$  [9](#). It represents the rate of change of an option's delta over time, often referred to as the "rho of delta." This is particularly important for traders who are concerned with how their delta-hedging needs will evolve as expiration approaches.

These formulas can be derived analytically from the Black-Scholes model and are available in academic resources and open-source libraries like `AmirDehkordi/OptionGreeks` [9](#). Their implementation follows the same pattern as GEX: the relevant chain data (prices, strikes, etc.) is pulled from DuckDB, passed as arrays to a Numba-compiled function decorated with `@numba.njit`, and a vectorized output of Vanna or Charm values for each strike is returned. This brings the platform's analytical capabilities to a level comparable to institutional-grade risk management systems. The combination of GEX, VPIN, and higher-order Greeks creates a multi-dimensional view of market structure, allowing the system to identify not only where dealers are structurally exposed but also how that exposure is changing dynamically due to shifts in price, time, and

volatility. This quantitative core, built on a foundation of rigorous mathematics and optimized for performance, is the engine that drives all subsequent analysis and visualization in Project Oracle.

| Metric                         | Formula / Definition                                                       | Significance                                                                                                                                 |
|--------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Gamma Exposure (GEX)           | $GEX = \sum (OI_i \times \Gamma_i \times \Delta_i \times 100 \times Spot)$ | Measures dealer hedging pressure. Positive GEX stabilizes markets; Negative GEX amplifies volatility <a href="#">75</a> <a href="#">97</a> . |
| Zero Gamma (ZG) Level          | Strike where cumulative net GEX flips sign.                                | Acts as a major structural support/resistance level determined by dealer positioning <a href="#">75</a> .                                    |
| Volume-Synchronized PIN (VPIN) | $VPIN = \frac{\sum}{\sum}$                                                 | $V^B_\tau - V^S_\tau$                                                                                                                        |
| Quote Imbalance (QI)           | $(BidSize - AskSize) / (BidSize + AskSize)$                                | Measures top-of-book LOB imbalance, used alongside VPIN to confirm directional flow toxicity <a href="#">126</a> .                           |
| Vanna                          | $\frac{d^2V}{dSd\sigma}$                                                   | Measures change in Delta with respect to Price or Vega with respect to Volatility; captures cross-gamma-vol risk <a href="#">9</a> .         |
| Charm                          | $\frac{d^2V}{dSdt}$                                                        | Measures the change in Delta over time ("rho of delta"); critical for theta decay and time-based hedging <a href="#">9</a> .                 |

## UI/UX Synthesis: Replicating Heatseeker and Flowseeker Visualizations

The ultimate utility of Project Oracle's sophisticated quantitative engine is realized through its user interface (UI) and user experience (UX). The goal of Phase 3 is to synthesize the real-time data and analytical outputs from the backend into intuitive, actionable visualizations that replicate the core functionality of established market tools like Skylit's Heatseeker and Flowseeker. This phase bridges the gap between abstract numerical results and concrete trader insight, requiring a focus on responsive design, real-time data streaming to the frontend, and the algorithmic detection of key market levels. The chosen technology stack of FastAPI for the backend and Dash with Plotly for the frontend provides a powerful and flexible framework for building these dynamic dashboards [3](#). Dash, in particular, excels at creating interactive web applications with Python, allowing for the direct integration of Plotly graphs, which are ideal for financial charting.

The centerpiece of the UI will be a "Heatseeker clone," a dynamic heatmap that visualizes Dealer Gamma Exposure (GEX) over time. This visualization will have the strike price on the Y-axis, time progressing along the X-axis, and the GEX value represented by a color



scale 75. In this representation, positive GEX (dealer stabilization) would be colored green (e.g., "Pika"), while negative GEX (dealer amplification) would be colored red ("Barney") 75. Superimposed on this heatmap will be a line graph representing the price of the underlying asset, plotted on a secondary Y-axis. This allows the user to directly correlate price action with the prevailing dealer hedging regime. A critical enhancement to this classic visualization is the algorithmic identification and highlighting of key structural levels. The system will scan the GEX data in real-time to detect "King Nodes," which are local maxima of absolute GEX values. These are points of maximum dealer hedging pressure and are often precursors to sharp reversals. Horizontal lines will be drawn on the heatmap at the strike prices of these King Nodes, labeled accordingly. Conversely, the system will also identify "Air Pockets," which are zones of near-zero GEX. These areas represent a lack of structural support or resistance from dealer hedging, suggesting that the market may be more susceptible to momentum-driven moves. Highlighting these nodes and pockets provides a powerful, at-a-glance summary of the market's structural integrity.

Complementing the heat map will be a "Flowseeker clone," implemented as a real-time scrolling ticker table. This component will display a live feed of significant trades, with rows being color-coded to indicate their potential impact on the market. The primary logic for this coloring will be based on the "Flow Scanner," a simple yet effective filter for identifying potentially anomalous trades. A trade will be flagged as high-risk if its size is greater than the remaining open interest (OI) for that option, or if its size exceeds the daily volume 75. Such trades could represent large block orders or attempts to manipulate the market. A bright yellow or red highlight will draw the user's attention to these events. The table might include columns for the option symbol, trade size, price, and the calculated color code. By presenting this information in a clean, uncluttered format, the Flowseeker clone provides traders with immediate feedback on the character of recent order flow.

Beyond these core replicants, the dashboard will feature a "Toxicity and Liquidity Gauge" widget. This compact widget will display two key real-time metrics: the VPIN CDF z-score and the Quote Imbalance (QI) z-score 29 126. The VPIN CDF z-score, as discussed previously, quantifies the extremity of order flow imbalance, while the Quote Imbalance measures the relative depth of the top of the book 126. The widget will trigger a strong visual alert (e.g., a flashing border or a change in background color) when a confluence of signals is detected. Specifically, an alert would activate when  $VPIN\ CDF > 0.5$  (indicating toxic flow) AND  $Quote\ Imbalance\ z-score > 1.5$  (indicating a severe imbalance in the top-of-book quotes) 29. This dual confirmation mechanism helps reduce false positives and focuses the trader's attention on moments of high informational



asymmetry and potential liquidity crisis. The entire dashboard will be built using Dash callbacks, which allow Python functions to update the `figure` and `children` properties of components in response to changes in other components or to timed intervals. This ensures that the UI remains fully synchronized with the backend analytics engine without requiring a full page reload.

Finally, to provide a richer context, the UI can be extended to incorporate real-time 3D visualizations of the implied volatility (IV) surface. By fetching the `iv` and other chain data from DuckDB, a PyTorch-based module can be used to render a mesh plot showing how IV varies with strike price and time to expiration [78](#). Integrating open-source implementations of Stochastic Volatility models like SABR or SVI could further enhance this visualization by providing a smoothed, interpolated surface that better reflects the true volatility landscape [23](#). This would require an additional research task to find and adapt a suitable open-source library for IV surface interpolation [68](#). The overall design philosophy for the UI/UX is centered on clarity, responsiveness, and actionability. Every element is designed to answer a specific question for the trader: Where are the structural supports and resistances? Is the flow of money informed or noise? Are there signs of liquidity stress? By synthesizing the complex backend analytics into these intuitive, real-time visualizations, Project Oracle can deliver a user experience that is not only competitive with existing commercial products but also customizable and extensible in ways that closed-source platforms cannot match.

## Advanced Applications: Developing an Institutional Edge with Novel Analytics

While replicating existing tools like Heatseeker and Flowseeker establishes a solid baseline, the true ambition of Project Oracle is to surpass them by developing novel, institutional-grade features that provide a unique analytical edge. Phase 4 of the project is dedicated to this goal, focusing on the creation of custom metrics and machine learning models that analyze dealer hedging behavior, inter-market relationships, and latent patterns in order flow. These advanced applications move the platform from a descriptive analytics tool to a prescriptive one, capable of identifying subtle market dynamics that are invisible to conventional retail-grade indicators. The development of these features requires a deep understanding of market microstructure theory and the application of cutting-edge techniques from computational physics and machine learning [16](#) [117](#).

One of the most powerful custom metrics to be developed is the **Trinity Alignment Index**. This index is designed to quantify the degree of structural alignment between the three most liquid equity indices: SPX (S&P 500 Index), SPY (S&P 500 ETF), and QQQ (Nasdaq 100 ETF). The core idea is that when dealers are simultaneously hedging large gamma exposures in the same relative position across these three markets, it signals a broad, systemic shift in market sentiment. The index will be calculated in real-time by tracking the "Zero Gamma" (ZG) strike levels for each of the three instruments [75](#). The ZG level represents the boundary between dealer-stabilizing and dealer-amplifying regimes [75](#). The system will compute the cross-correlation between the time series of these ZG levels. A high correlation would indicate that the ZG levels are moving in tandem, suggesting synchronized dealer positioning. The final output will be a normalized score from 0 to 100, where a high score denotes a high degree of alignment and a potential precursor to a broad market turn. This metric directly addresses the need for signal coherence across different but related markets, a concept highlighted in systematic trading frameworks [58](#).

Another innovative feature is the **Node Lifecycle Tracker**. This component adds a historical and dynamic dimension to the static GEX heatmap. The tracker implements a state machine that monitors how many times the price of the underlying asset has "tapped" or crossed a "King Node"—a local maximum of absolute GEX [75](#). Each time the spot price touches such a node, the node's "lifecycle counter" is incremented. More importantly, its visual representation on the heatmap decays over time. On each successive tap, the node's opacity is reduced, and its contribution to the overall GEX aggregation is diminished. This visual metaphor communicates the concept of "structural fatigue." A node that has held the market off for an extended period becomes weaker with each subsequent test, signaling increased risk of a break. This feature provides traders with a dynamic sense of which structural levels are currently "live" and which have lost their potency, adding a crucial temporal dimension to the otherwise static picture of dealer exposure.

Perhaps the most technologically advanced feature is the implementation of an **Anomaly Detection System** using a 1D-CNN Autoencoder. Traditional methods for detecting "toxic flow," like VPIN thresholds, are reactive and rely on pre-defined statistical assumptions. An autoencoder-based approach offers a more adaptive and powerful alternative. The system will ingest multivariate time series data, primarily the VPIN readings and the Quote Imbalance values, as input sequences [109](#)[110](#). A 1D-CNN Autoencoder is trained on a large dataset of "normal" market conditions, learning to reconstruct the input time series with minimal error [111](#). During training, the model identifies the underlying patterns and correlations in the data. Once trained, it is deployed to run in inference

mode. When presented with new data, if the reconstruction error—the difference between the input and the output—is significantly larger than the error observed during training, it is flagged as an anomaly [113141](#). This method can detect novel forms of toxic flow that do not conform to traditional patterns, acting as a real-time alarm for unexpected market behavior. The Hugging Face directive ensures that the necessary infrastructure is in place to acquire state-of-the-art transformer and autoencoder models and datasets like FI-2010 for training and validation [78 111](#). Other advanced research areas to explore include modeling the Limit Order Book (LOB) itself using Hawkes Processes, which are stochastic models that describe the clustering of events like trades and cancellations, and using Transformer-based architectures like LiT to forecast short-term market movements directly from LOB data [16 106117](#).

By integrating these advanced applications, Project Oracle transcends the role of a mere data viewer. It becomes an active participant in the analytical process, surfacing insights that require a synthesis of microstructure theory, quantitative modeling, and machine learning. The Trinity Alignment Index provides a macro perspective on market structure, the Node Lifecycle Tracker adds a crucial historical context to key levels, and the Anomaly Detection system offers a proactive defense against unforeseen market shocks. Together, these features constitute the "institutional edge" that was a core objective of the project, empowering users with a level of analytical depth previously accessible only through expensive, proprietary platforms.

| Feature                           | Description                                                                                                                                       | Underlying Technology / Theory                                                                      |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Trinity Alignment Index           | A 0-100 score measuring the cross-correlation of Zero-Gamma (ZG) strike levels across SPX, SPY, and QQQ to gauge systemic dealer positioning.     | Cross-correlation of time series, Market Structure Analysis <a href="#">58 75</a>                   |
| Node Lifecycle Tracker            | A state machine that tracks taps on King Nodes (high GEX levels) and decays their visual weight and importance on successive tests.               | State Machine, Market Memory / Structural Fatigue Concept <a href="#">75</a>                        |
| Anomaly Detection (Autoencoder)   | A 1D-CNN Autoencoder trained to reconstruct normal VPIN and Quote Imbalance time series, flagging significant reconstruction errors as anomalies. | Unsupervised Machine Learning, Reconstruction Error, Time Series Analysis <a href="#">109110111</a> |
| Limit Order Book (LOB) Simulation | Simulates realistic LOB dynamics using event-based models, potentially incorporating Hawkes Processes for trade arrival.                          | Event-Based Simulation, Stochastic Processes, Reinforcement Learning <a href="#">16 106 115</a>     |
| Stochastic Volatility Surface     | Renders a 3D mesh of the Implied Volatility (IV) surface, potentially using models like SABR or SVI for smoothing and interpolation.              | Stochastic Volatility Models, Surface Interpolation, 3D Visualization <a href="#">23 117</a>        |

# Autonomous Research & Asset Acquisition Protocol

A defining characteristic of Project Oracle is its mandate to evolve and improve through continuous, autonomous discovery and integration of cutting-edge knowledge. This protocol elevates the project from a static implementation of known formulas to a living research platform that stays abreast of the latest developments in quantitative finance. The directive granted to the AI agent includes explicit instructions to leverage its web search, GitHub scraping, and code analysis capabilities to proactively seek out, study, and adapt state-of-the-art open-source code and academic research [23](#). This self-improving capability is the key to maintaining the platform's "institutional edge" over time. The research mandate is focused on several key frontiers of market microstructure and algorithmic trading: Limit Order Book (LOB) dynamics, Hawkes Processes for modeling trade arrivals, advanced Stochastic Volatility models, and next-generation machine learning for flow toxicity and anomaly detection [16](#) [18](#) [117](#).

The first step in this protocol is the active discovery of relevant repositories and papers. The agent is instructed to search platforms like GitHub, arXiv, and QuantConnect for projects related to the aforementioned topics. For instance, regarding LOB dynamics, the agent would search for implementations of event-based simulators or deep learning models like the Limit Order Book Transformer (LiT) [117](#)[118](#). For Hawkes Processes, it would look for libraries like `HawkesPyLib` or `hawkeslib` that facilitate simulation and parameter estimation [13](#) [14](#). Upon discovering a promising repository, the agent must perform a deep study of its methodology, extracting the core mathematical kernels rather than attempting to copy-paste the entire codebase [23](#). This extracted logic must then be translated into production-grade, Numba-optimized Python that adheres to the project's strict architectural constraints. For example, finding a research paper on a novel Hawkes Process model would lead to the extraction of the likelihood function and optimization routines, which would then be adapted for use in the VPIN and LOB engine modules.

The second critical component of the protocol is the acquisition and organization of state-of-the-art machine learning assets. The project's advanced applications, particularly the anomaly detector, depend heavily on having access to pre-trained models and high-quality datasets. The Hugging Face ecosystem is central to this effort. The agent is tasked with provisioning the complete PyTorch/Hugging Face stack, including `transformers`, `datasets`, and `accelerate` [71](#). It will then execute a script to search the Hugging Face Hub for specific types of assets. This includes searching for: 1. **Time-Series Forecasting Models:** SOTA models like PatchTST, Informer, or Autoformer that are designed for forecasting tasks on sequential data [34](#) [36](#). 2. **Anomaly Detection Models:** 1D-CNN Autoencoders or Transformer-based architectures specifically used for financial

anomaly detection [111130131](#). 3. **Limit Order Book Prediction Models:** Specialized models like DeepLOB or LiT that are trained on high-frequency LOB data [31 117](#).

Simultaneously, the agent will search for relevant datasets, such as the FI-2010 benchmark dataset or NASDAQ LOB snapshots, which are invaluable for pre-training and validating the deep learning models [40 78](#). All downloaded models and datasets must be organized into a strict local directory structure: `./project_oracle/models/` and `./project_oracle/datasets/` [78](#). This systematic approach to asset acquisition ensures that the project has access to the best available tools and data, completely bypassing the need for manual scraping and cleaning of messy raw data. The use of GPU-accelerated simulators like JAX-LOB, which can process thousands of books in parallel, further underscores the commitment to leveraging high-performance computing for research [17 105](#).

This autonomous research protocol is not just a one-time setup but a continuous process. As the project matures, the agent's search parameters can be refined, and it can be tasked with monitoring specific researchers or repositories for new publications and code updates. This ensures that Project Oracle does not become a static artifact but a dynamic platform that grows in sophistication over time. The integration of findings from academic papers, such as those examining the relationship between order-flow toxicity and price jumps in Bitcoin or the effect of monetary policy on traded value, can provide novel inputs for the analytical engine [21 92](#). The ultimate goal is to create a virtuous cycle: the system generates analytical signals, researchers analyze the effectiveness of these signals, and the system is automatically updated with improved models and methodologies discovered through this research. This approach embodies the spirit of open science and collaborative intelligence, using the power of automation to accelerate the pace of quantitative discovery. By combining rigorous engineering with an agile, research-driven development process, Project Oracle is positioned to become a truly unique and powerful tool in the field of options analytics.

| Asset Type               | Target Examples                                                                       | Purpose in Project Oracle                                                                            |
|--------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Time-Series Models       | PatchTST, Informer, Autoformer <a href="#">34</a> <a href="#">36</a>                  | Forecasting VPIN and Quote Imbalance time series to predict future flow toxicity.                    |
| Anomaly Detection Models | 1D-CNN Autoencoders, Transformer-based GANs <a href="#">11130131</a>                  | Training unsupervised models to detect novel "toxic flow" anomalies deviating from learned patterns. |
| LOB Prediction Models    | LiT, DeepLOB, Neural Hawkes Process <a href="#">31</a> <a href="#">106117</a>         | Modeling and predicting Limit Order Book dynamics for advanced microstructure analysis.              |
| Datasets                 | FI-2010, NASDAQ LOB, High-Frequency Crypto Data <a href="#">40</a> <a href="#">78</a> | Providing benchmark data for training and validating deep learning models on realistic order flow.   |
| Volatility Models        | SABR, SVI, Rough Bergomi <a href="#">23</a> <a href="#">85</a>                        | Improving the accuracy of the Implied Volatility (IV) surface used for Greek calculations.           |

## Execution Strategy and Phased Implementation Roadmap

The successful realization of Project Oracle requires a disciplined, phased execution strategy that builds the system's capabilities incrementally, starting with a robust foundation and progressively layering on complexity. The roadmap is divided into four distinct phases, each with a clear objective and a set of deliverables. This structured approach ensures that each component is developed, tested, and integrated systematically, mitigating risk and providing tangible progress milestones. The entire process is designed to be executed by an autonomous AI agent, guided by the master directive provided in the conversation history, with the GitHub Student Developer Pack serving as the enabler for version control, CI/CD, and deployment [70](#) [71](#).

**Phase 1: Foundational Infrastructure & Data Pipeline.** The primary goal of this phase is to establish a reliable, real-time data ingestion and storage pipeline. Without this bedrock, no subsequent analysis can occur. The first action is to scaffold the monorepo with the designated directory structure: `/ingestion`, `/analytics`, `/api`, `/ui`, and `/research` [74](#). Following this, the core task is to implement the WebSocket client. Leveraging the logic from the `tylerebowers/Schwabdev` repository, the agent will write a Python class to manage the connection to the broker's streaming API, subscribe to Level 2 options and underlying asset ticks, and parse the incoming JSON payloads [11](#). Concurrently, the agent must define the DuckDB schema. This involves writing the SQL statements to create the `ticks`, `chains`, and `lob_snapshots` tables with appropriate data types [3](#). The final piece of Phase 1 is the implementation of the async batching

writer. This will involve creating a bounded `asyncio.Queue` and a dedicated coroutine that consumes items from the queue, batches them, and executes bulk `INSERT` statements into the DuckDB tables every 50 milliseconds [122](#). The success criterion for this phase is a running script that can connect to the WebSocket feed, receive data, and see it appear reliably in the DuckDB tables.

**Phase 2: The Mathematical Engine & Continuous Research.** With a stable data pipeline in place, Phase 2 focuses on building the core analytical engine. The first task is to implement the Greek calculator. Using the analytical formulas for Vanna and Charm from the `AmirDehkordi/OptionGreeks` repository, the agent will write Numba-compiled (`@numba.njit`) functions that accept NumPy arrays of chain data and return vectorized outputs of Vanna and Charm [9](#). Next, the GEX/VEX aggregator will be implemented as another Numba function that queries the current chain state from DuckDB and computes the aggregate GEX array. Following this, the VPIN engine will be built, adapting the volume-bucketing and Bulk Volume Classification logic from repositories like `theopenstreet/VPIN_HFT` [68](#). This function must maintain a rolling state in memory to process the continuous stream of trade data. Immediately following these core implementations, the autonomous research mandate is triggered. The agent will execute its first research task: to find and summarize three advanced GitHub repositories related to Limit Order Book (LOB) imbalance predictive modeling and Hawkes Processes for trade arrival [16](#) [18](#). Based on this research, it will begin drafting a Numba-optimized module for the `/analytics` folder. The success criterion for Phase 2 is a set of working, Numba-compiled functions that can take raw tick data as input and produce GEX, VPIN, and higher-order Greeks as real-time output.

**Phase 3: Synthesis & UI (Dash/Plotly).** This phase is about translating the analytical output into a usable and insightful user interface. The first deliverable is the Heatseeker clone. Using Dash and Plotly, the agent will create a server-side rendered heatmap where the Y-axis is strike price, the X-axis is time, and the color scale represents GEX [3](#). The underlying spot price will be overlaid as a secondary y-axis line. Algorithmic functions will be written to detect and draw horizontal lines for "King Nodes" (local GEX maxima) and shade "Air Pockets" (zones of near-zero GEX). The second deliverable is the Flowseeker clone, a real-time scrolling DataTable that color-codes rows based on the "Flow Scanner" logic (`Trade_Size > 0I`) [75](#). The final UI component is the Toxicity and Liquidity Dashboard, a small widget displaying the VPIN CDF z-score and Quote Imbalance, with a callback that triggers a visual alert when both are above specified thresholds [29](#). Before completing the UI, the agent's second research task is initiated: to find the best open-source implementations of Stochastic Volatility Surface interpolation (e.g., SABR) and integrate the math into the UI to render a real-time 3D IV surface mesh



23 . The success criterion is a fully functional, single-page web application that displays all synthesized data visually.

**Phase 4: Advanced Applications & Institutional Edge.** The final phase is dedicated to building the platform's unique, advanced features. The first task is to implement the Trinity Alignment Index by cross-correlating the Zero-Gamma strike time series of SPX, SPY, and QQQ 58 . The second is the Node Lifecycle Tracker, a state machine that manages the tapping and decay of King Nodes on the heatmap 75 . The third and most complex task is the Anomaly Detection system. This involves scaffolding a PyTorch neural network module to ingest the VPIN and Quote Imbalance time series. The agent will then execute the Hugging Face asset acquisition script to download a suitable 1D-CNN Autoencoder model and a benchmark dataset like FI-2010 for pre-training 78 111. The final research task is to investigate and attempt to implement Kyle's Lambda and Amihud Illiquidity metrics to generate a real-time "Market Fragility" score 28 . The success criterion for Phase 4 is a fully operational, self-hosted analytics terminal that not only replicates but demonstrably extends the capabilities of existing retail tools, embodying the "institutional edge" envisioned in the project's core objective. Throughout this roadmap, GitHub Actions will be configured for automated testing and linting, and the final product will be containerized using Docker and pushed to the GitHub Container Registry (GHCR) for easy deployment on a personal VPS 70 71 .

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